Week 10: Monday, AST 5011: Astrophysical Systems

The Galaxy-Halo Connection & Gas Accretion

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Reference: Cimatti, Fraternali & Nipoti, Ch. 5-6

With material from Benedikt Diemer (UMD) and Frank van den Bosch (Yale).

The Stellar Mass Function

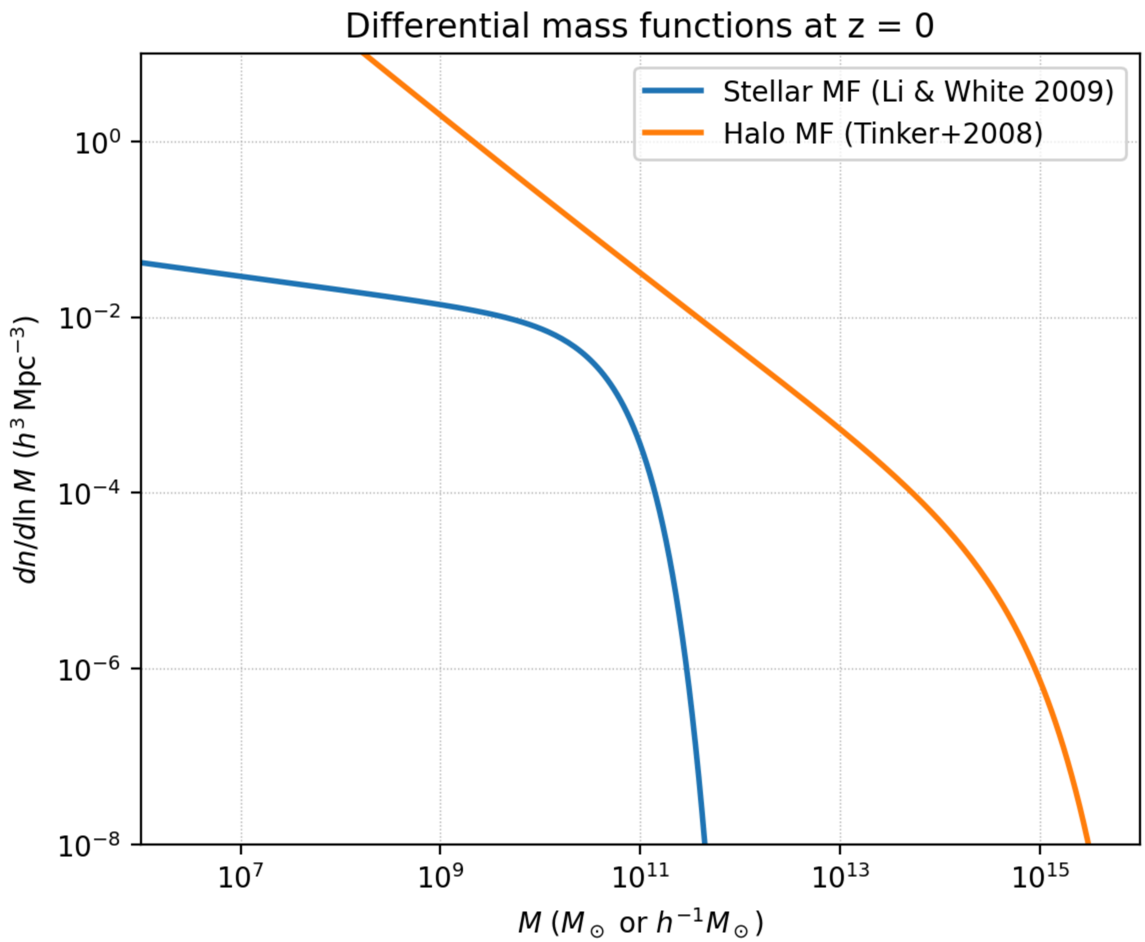
The stellar mass function (SMF) describes the number density of galaxies as a function of their stellar mass. It is well fit by a Schechter function:

$$\phi(M)\, d\ln M = \phi^* \left(\frac{M}{M^*}\right)^{\alpha+1} \exp\left(-\frac{M}{M^*}\right) d\ln M$$

where:

- ϕ^* is the normalization (number density at the "knee")
- M^* is the characteristic mass where the exponential cutoff begins
- α is the faint-end slope ($\alpha < -1$ means many more faint galaxies than bright ones)

We use the fit from Li & White (2009) based on SDSS data: $M^* = 10^{10.525}\, M_\odot$, $\alpha = -1.155$, $\phi^* = 0.0083\, h^3\, \mathrm{Mpc}^{-3}$



Abundance Matching

The fundamental idea of abundance matching is simple:

> Rank-order galaxies by stellar mass and halos by halo mass. The most massive galaxy lives in the most massive halo.

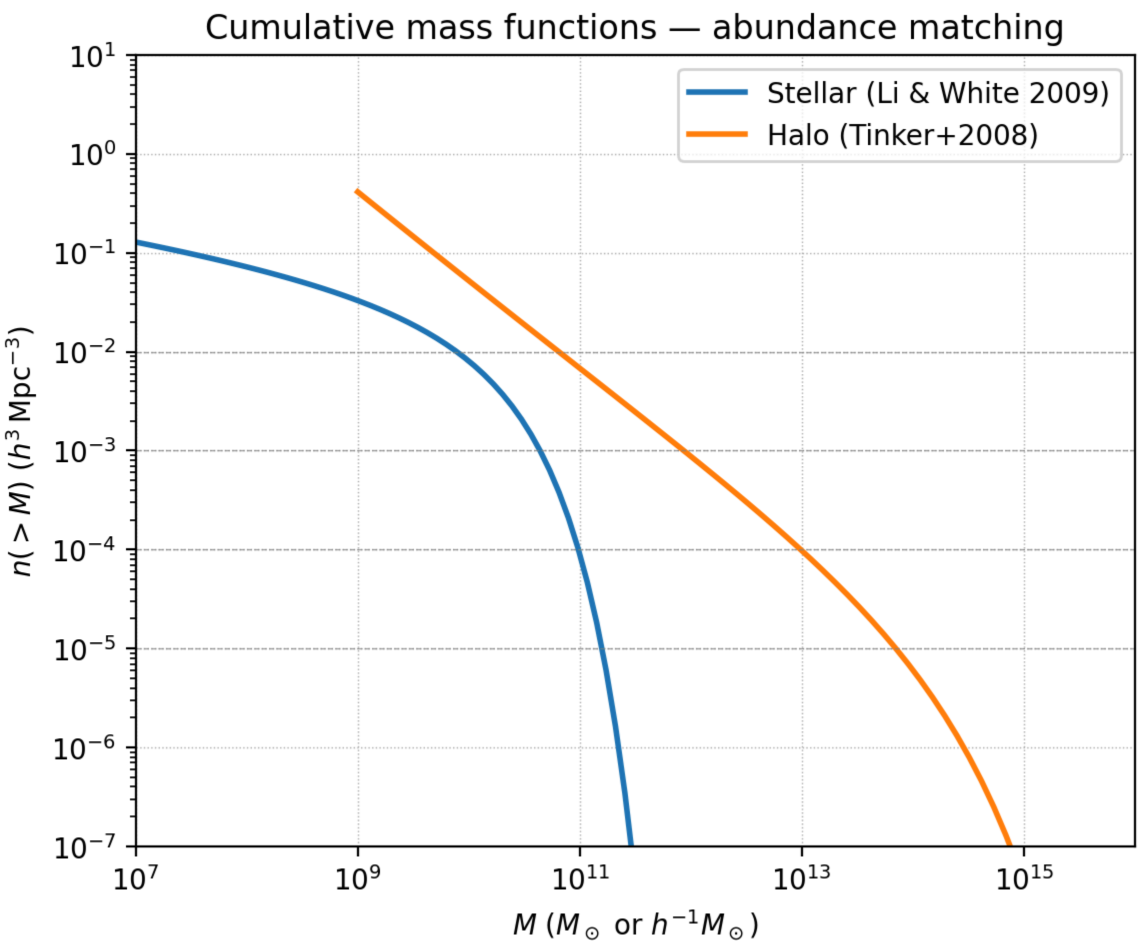
Formally, we match at fixed cumulative number density:

$$n(> M_*) = n(> M_{\rm halo})$$

This requires computing the cumulative (integral) mass functions from high mass downward. The matching then gives us a one-to-one mapping $M_*(M_{\rm halo})$ — the stellar-to-halo mass relation (SHMR).

This simple model works remarkably well because both mass functions are monotonically decreasing.

```
Computing cumulative stellar mass function...
Computing cumulative halo mass function...
Done.
```



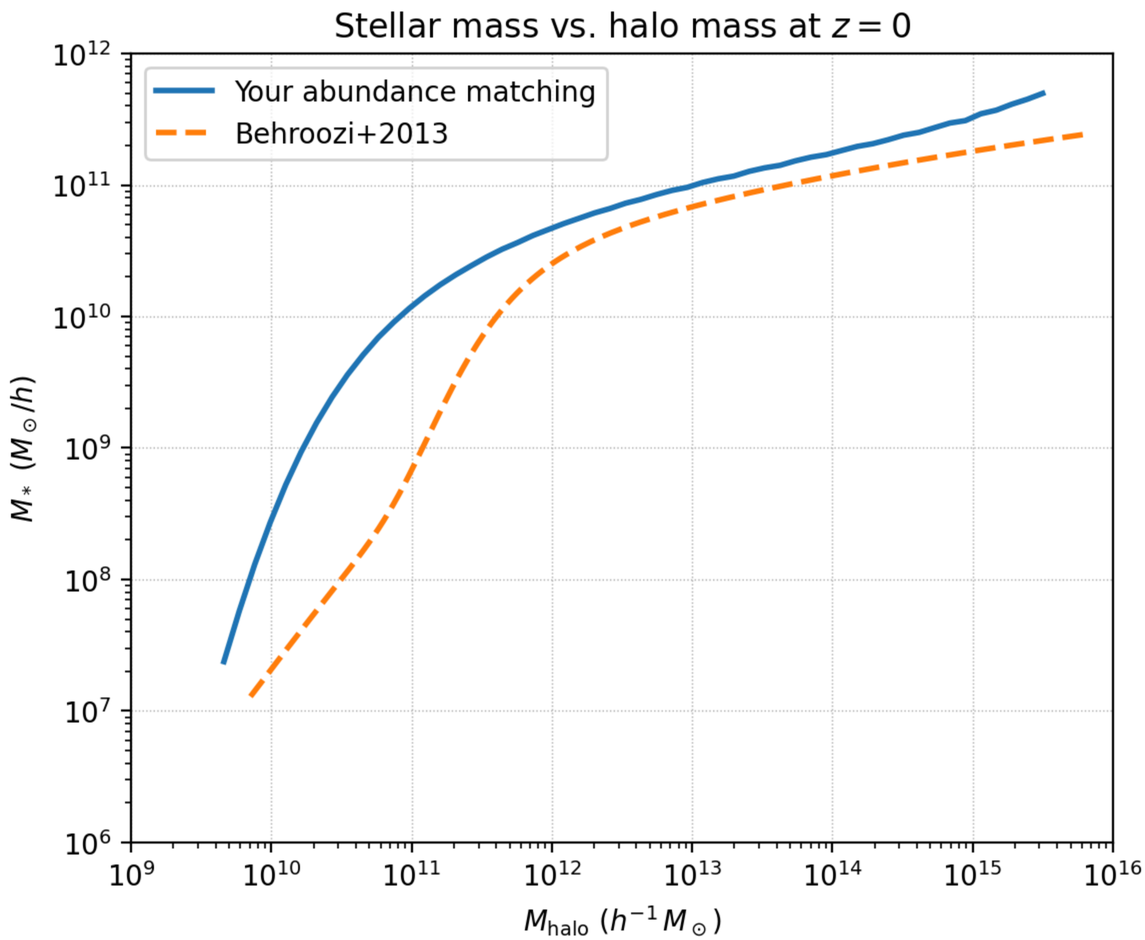
Exercise 1: Abundance Matching

In this exercise, you will perform abundance matching to derive the stellar-to-halo mass relation $M_*(M_{\rm halo})$.

Steps:

1. The cumulative mass functions have been pre-computed above.
2. FILL IN: For each halo mass, find the stellar mass at the same cumulative number density using interpolation.
3. Plot the resulting $M_*(M_{\rm halo})$ relation.
4. Compare to the Behroozi et al. (2013) observational constraint.

Hint: Use `np.interp()` to interpolate. Since `np.interp` requires monotonically increasing x, you may need to flip the arrays.



Abundance Matching with Simulated Halos

The analytical approach above uses the Tinker+2008 halo mass function, which counts only distinct (host) halos. But the stellar mass function includes satellite galaxies living in subhalos. To properly match, we should include subhalos in the halo count.

We use halo catalogs from the Erebos N-body simulations, where we track the peak mass $M_{\rm peak}$ — the maximum virial mass a halo ever attained. This is a better proxy for galaxy stellar mass because subhalos lose mass after infall (tidal stripping), but their galaxies largely retain their stars.

The Stellar-to-Halo Mass Relation (SHMR)

The SHMR is often expressed as the ratio $M_*/M_{\rm halo}$, which reveals the star formation efficiency — what fraction of the available baryons have been converted to stars.

Key features:

- The ratio peaks at $M_{\rm halo} \sim 10^{12}\,M_\odot$ (Milky Way-mass halos), where $M_*/M_{\rm halo} \sim 0.03$
- At lower masses, feedback from supernovae and stellar winds suppresses star formation

- At higher masses, AGN feedback heats the gas and shuts down star formation
- No halo converts more than $\sim 20\%$ of its baryons into stars

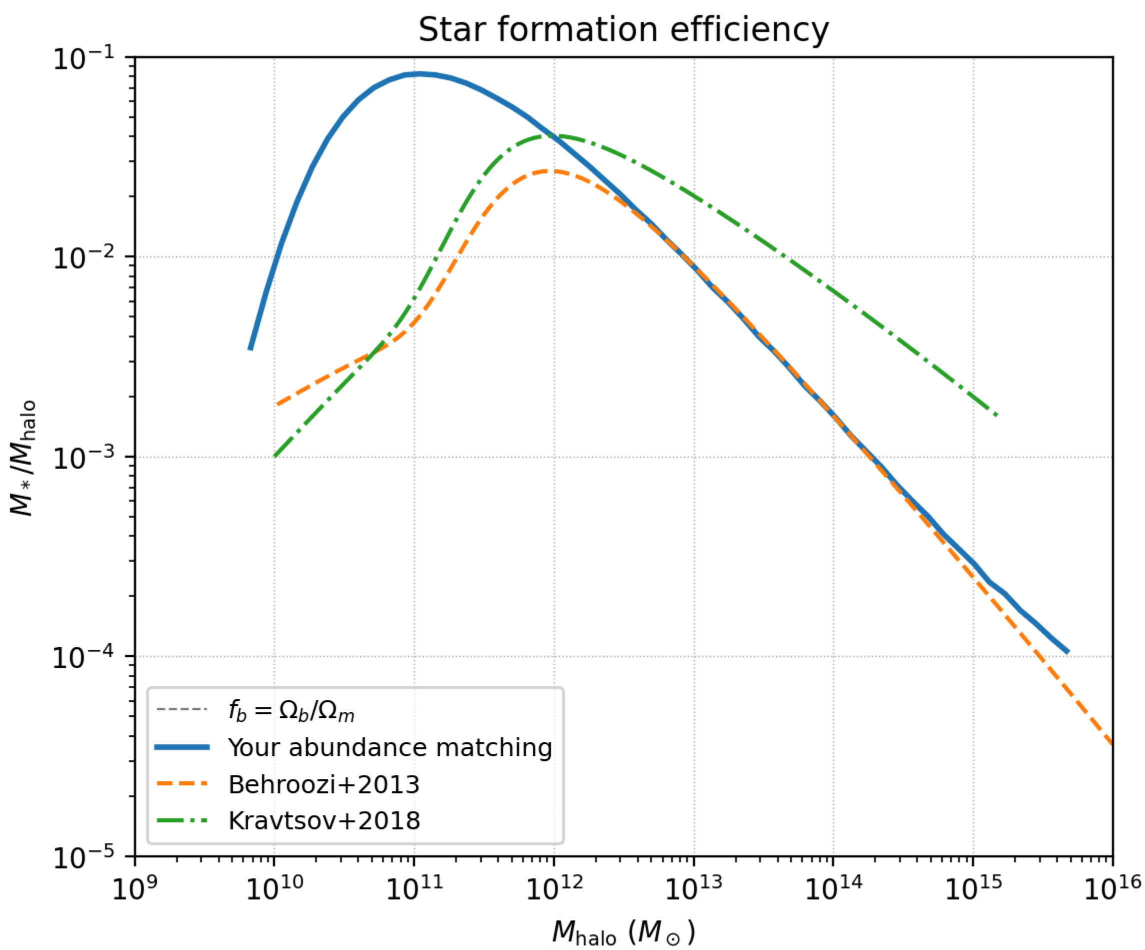
Exercise 2: Star Formation Efficiency

In this exercise, you will compute the star formation efficiency as a function of halo mass and compare to observational constraints.

Steps:

1. FILL IN: Compute the ratio $M_*/M_{\rm halo}$ from your abundance matching result
2. FILL IN: Compute the cosmic baryon fraction $f_b = \Omega_b/\Omega_m$
3. FILL IN: Compute the star formation efficiency $\epsilon = (M_*/M_{\rm halo})/f_b$
4. Compare to the Behroozi+2013 and Kravtsov+2018 observational constraints
5. Find the halo mass where efficiency peaks

```
Cosmic baryon fraction f_b = 0.1575
M_halo = 1e+11 Msun: M*/M_halo = 0.0822, efficiency = 52.2%
M_halo = 1e+12 Msun: M*/M_halo = 0.0382, efficiency = 24.2%
M_halo = 1e+13 Msun: M*/M_halo = 0.0086, efficiency = 5.4%
M_halo = 1e+14 Msun: M*/M_halo = 0.0016, efficiency = 1.0%
```



SHMR Evolution with Redshift

The stellar-to-halo mass relation evolves with cosmic time. Using the redshift-dependent fitting function from Behroozi et al. (2013), we can track how the peak efficiency mass and the overall normalization shift. At higher redshifts, halos of a given mass have had less time to form stars, so the efficiency is systematically lower.

Galaxy Bias

Galaxies do not perfectly trace the underlying dark matter distribution. They are biased tracers: the galaxy overdensity is related to the matter overdensity by a bias factor b:

$$\delta_g(\mathbf{x}) = b \cdot \delta_m(\mathbf{x})$$

This means the galaxy power spectrum is:

$$P_{gg}(k) = b^2 \cdot P_{mm}(k)$$

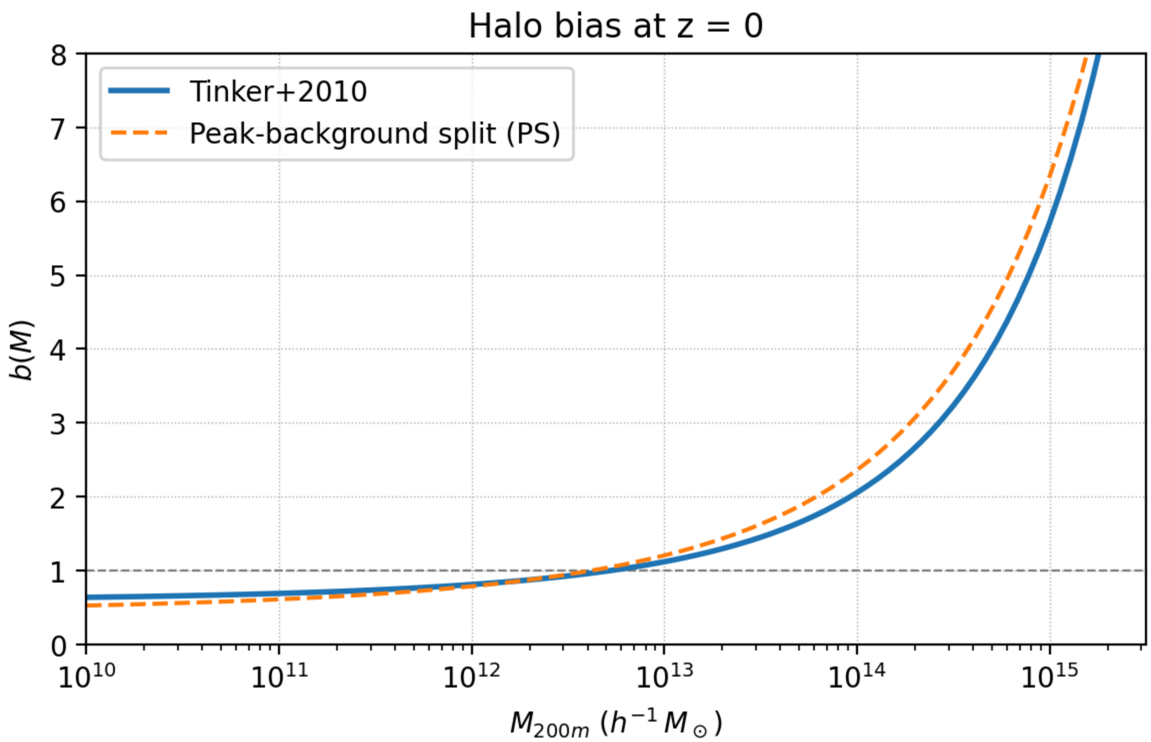
The bias depends on halo mass:

- Low-mass halos ($M \lesssim 10^{13}\, M_\odot$) are anti-biased ($b < 1$): they live in average environments
- Massive halos ($M \gtrsim 10^{13}\, M_\odot$) are positively biased ($b > 1$): they preferentially form in overdense regions

The peak-background split model predicts:

$$b(\nu) = 1 + \frac{\nu^2 - 1}{\delta_c}$$

where $\nu = \delta_c / \sigma(M)$ is the peak height. More sophisticated models (Tinker+2010) calibrate this against simulations.




```
b(M = 1e+11 h^-1 Msun) = 0.69
b(M = 1e+12 h^-1 Msun) = 0.81
b(M = 1e+13 h^-1 Msun) = 1.12
b(M = 1e+14 h^-1 Msun) = 2.05
b(M = 1e+15 h^-1 Msun) = 5.71
```

Bias Evolution with Redshift

At higher redshifts, a halo of a given mass is rarer (a more extreme fluctuation), and therefore more biased. This has important implications for galaxy surveys: the same galaxy population traces the matter field more strongly at earlier times.

Effective Galaxy Bias

In practice, a galaxy sample includes halos of many different masses. The effective bias of the sample is the mass-function-weighted average:

$$b_{\text{eff}} = \frac{\int_{M_{\min}}^{\infty} b(M)\, \frac{dn}{d\ln M}\, d\ln M}{\int_{M_{\min}}^{\infty} \frac{dn}{d\ln M}\, d\ln M}$$

For a magnitude-limited sample, $M_{\min}$ corresponds to the minimum halo mass that hosts a galaxy bright enough to be observed.

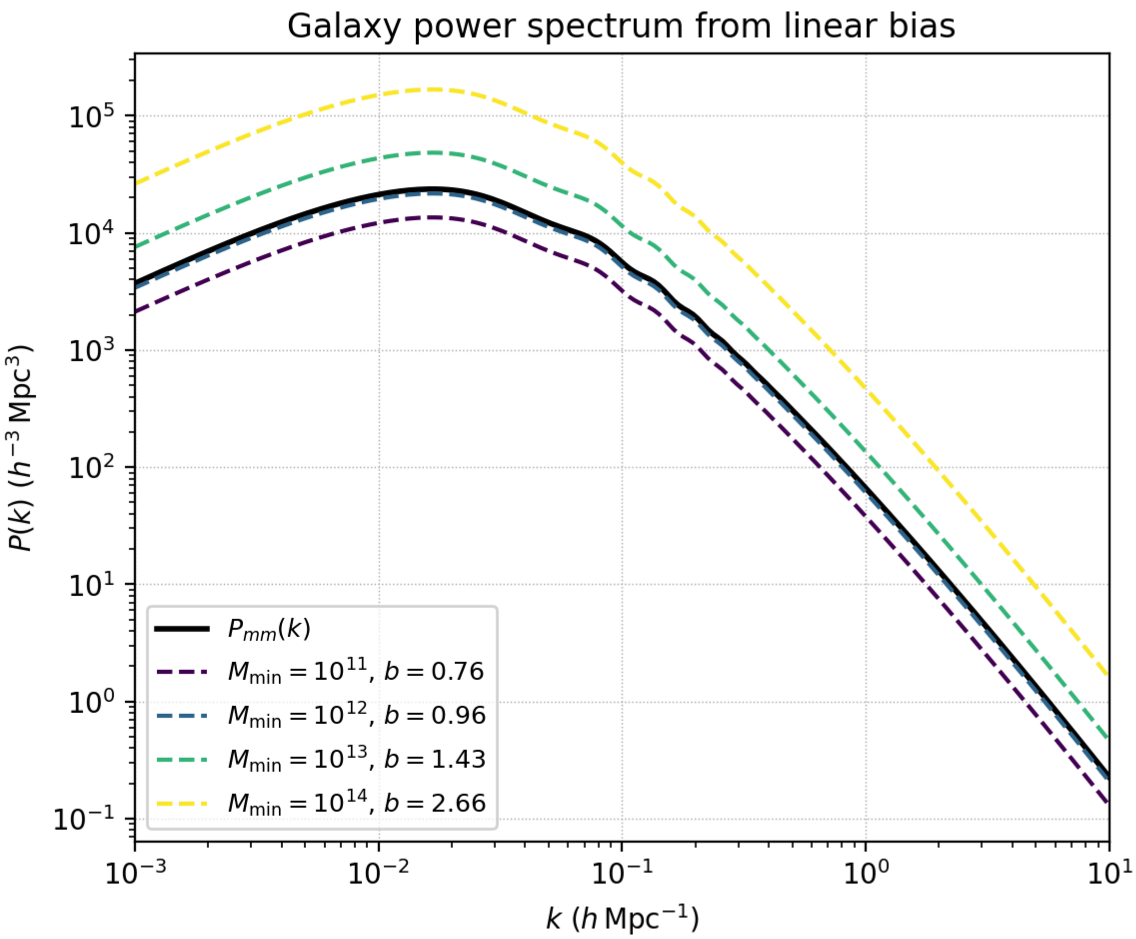
Exercise 3: Galaxy Bias

In this exercise, you will compute the effective galaxy bias for different halo mass thresholds and predict the galaxy power spectrum.

Steps:

1. FILL IN: Compute the halo bias $b(M)$ using `bias.haloBias()`
2. FILL IN: Compute the effective bias $b_{\rm eff}$ by weighting with the mass function
3. FILL IN: Predict the galaxy power spectrum $P_{gg}(k) = b_{\rm eff}^2 \cdot P_{mm}(k)$
4. Plot $P_{gg}(k)$ for different mass thresholds and compare to $P_{mm}(k)$

```
M_min = 1e+11: b_eff = 0.76, n_gal = 3.44e-02 h^3/Mpc^3
M_min = 1e+12: b_eff = 0.96, n_gal = 4.47e-03 h^3/Mpc^3
M_min = 1e+13: b_eff = 1.43, n_gal = 5.00e-04 h^3/Mpc^3
M_min = 1e+14: b_eff = 2.66, n_gal = 3.11e-05 h^3/Mpc^3
```



Part 2: Gas Accretion & Cooling

Reference: Cimatti, Fraternali & Nipoti, Ch. 6

With material from Benedikt Diemer (UMD) and Frank van den Bosch (Yale).

Accretion Shocks and the Virial TemperatureWhen gas accretes onto a dark matter halo, it encounters an accretion shock near the virial radius, heating the gas to approximately the virial temperature:

$$T_{\rm vir} = \frac{\mu m_p}{3k_B} V_{\rm vir}^2$$

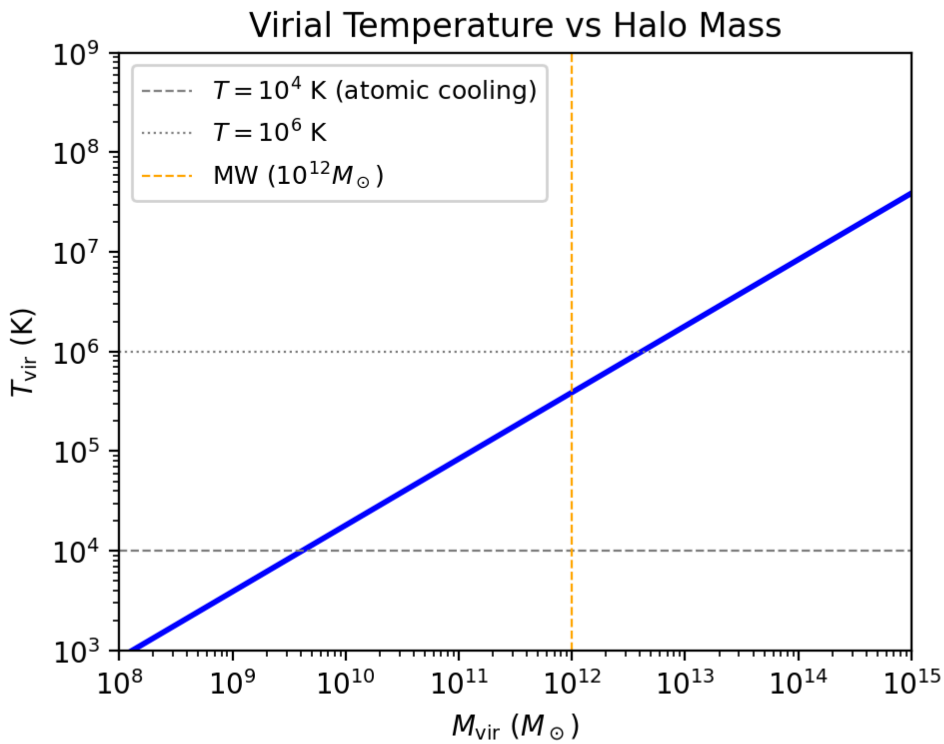
where $\mu \approx 0.59$ is the mean molecular weight for a fully ionized primordial gas, and $V_{\rm vir}$ is the circular velocity at the virial radius.The virial velocity scales as $V_{\rm vir} \propto M^{1/3}$, so $T_{\rm vir} \propto M^{2/3}$.For a Milky Way-mass halo ($M_{\rm vir} = 10^{12}\, M_\odot$), the virial temperature is about 4×10^5 K.

```
MW halo: V_vir = 127.7 km/s
MW halo: T_vir = 3.89e+05 K

T_vir = 2.38e+05 K * (V_vir/100 km/s)^2
```

Exercise 4: Virial Temperature vs. Halo MassCompute the virial temperature as a function of halo mass from 10^8 to $10^{15}\, M_\odot$. Identify the mass scales corresponding to:- Atomic hydrogen cooling limit: $T = 10^4$ K- Milky Way halo: $M = 10^{12}\, M_\odot$- Galaxy cluster: $M = 10^{14.5}\, M_\odot$Also overplot lines marking key temperature thresholds.

```
Atomic cooling limit (T=10^4 K): M_halo = 4.12e+09 Msun
MW halo (10^12 Msun): T_vir = 3.89e+05 K
Cluster (10^14.5 Msun): T_vir = 1.80e+07 K
```



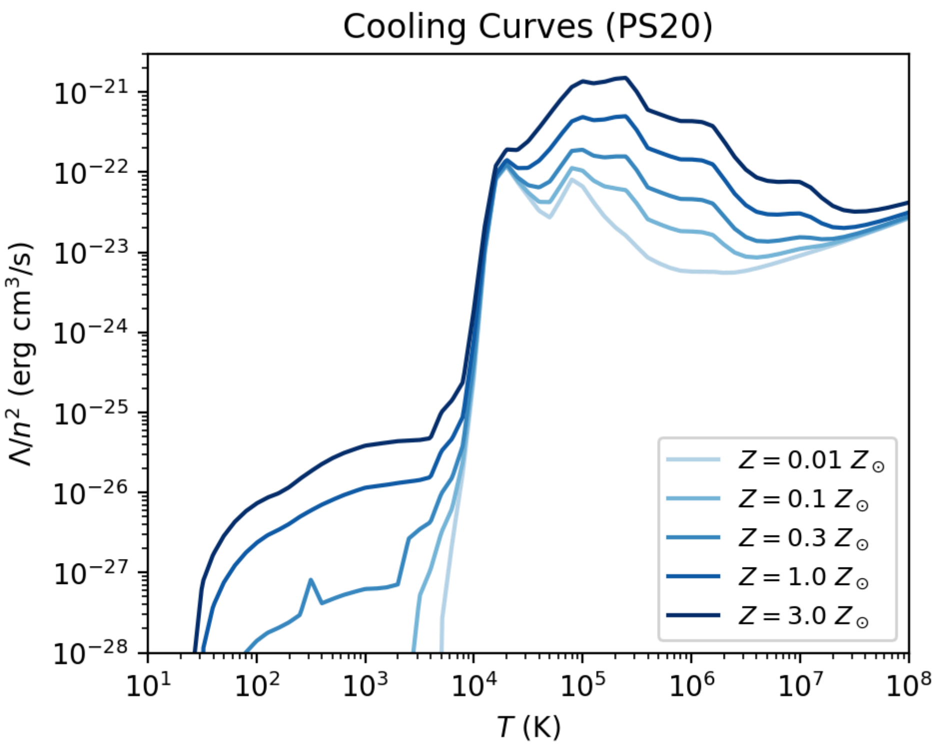
Demonstration: Virial Temperature at Multiple Redshifts

The virial temperature depends on redshift because $V_{\rm vir}^2 \propto M/R_{\rm vir}$, and the virial radius depends on the background density $\rho_{\rm vir}(z)$, which increases with redshift.

At fixed halo mass, $T_{\rm vir}$ is **higher** at higher redshift because the halo is more compact ($R_{\rm vir}$ is smaller). This means that massive halos at high z can reach very high temperatures, while the atomic cooling floor ($T = 10^4$ K) corresponds to lower halo masses at high z.

Cooling MechanismsThe post-shock gas must cool in order to condense and form galaxies. The main cooling mechanisms are:1. Bremsstrahlung (free-free): dominant at $T > 10^7$ K. $\Lambda \propto T^{1/2}$.2. Recombination + line emission: dominant at $10^4 < T < 10^7$ K. Metal lines (C, N, O, Fe) greatly enhance cooling.3. Compton cooling: scattering off CMB photons, important at high redshift.4. Molecular cooling: H_2 and HD at $T < 10^4$ K (important for first stars).The net cooling rate $\Lambda(T, n, Z)$ depends on temperature, density, and metallicity. We use the tables of Plöckinger & Schaye (2020), which include both cooling and heating (e.g., photoheating from the UV background).The

quantity Λ/n^2 (erg cm^3 s^{-1}) is approximately density-independent at high T.



Demonstration: Density Dependence of Cooling

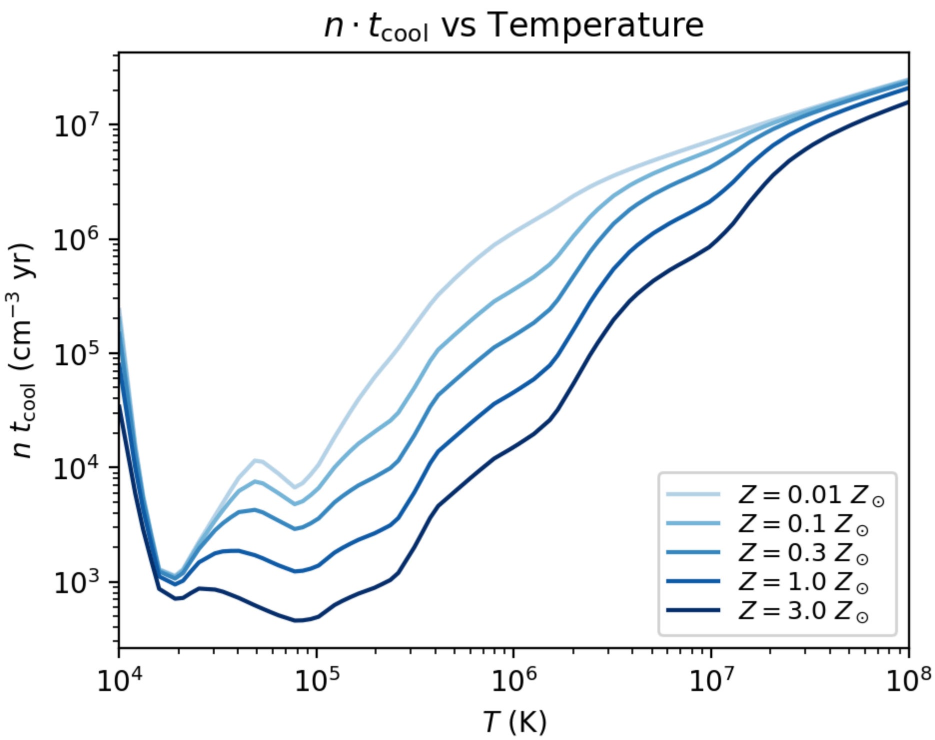
The plot above shows Λ/n^2 at different metallicities but fixed density $n = 1\,\mathrm{cm}^{-3}$. However, the cooling rate is **not** perfectly density-independent: at low temperatures ($T \lesssim 10^4$ K), photoheating from the UV background competes with radiative cooling, and the net rate depends on density.

The figure below shows Λ/n^2 at different gas densities for fixed solar metallicity. At high T, the curves converge (bremsstrahlung and line cooling dominate), but at low T they diverge significantly due to density-dependent photoionization equilibrium (adapted from CFN Chapter 6).

The Cooling TimeThe cooling time is the time for the gas to radiate away its thermal energy:

$$t_{\rm cool} = \frac{3}{2}\frac{nk_BT}{\Lambda(T,n,Z)}$$

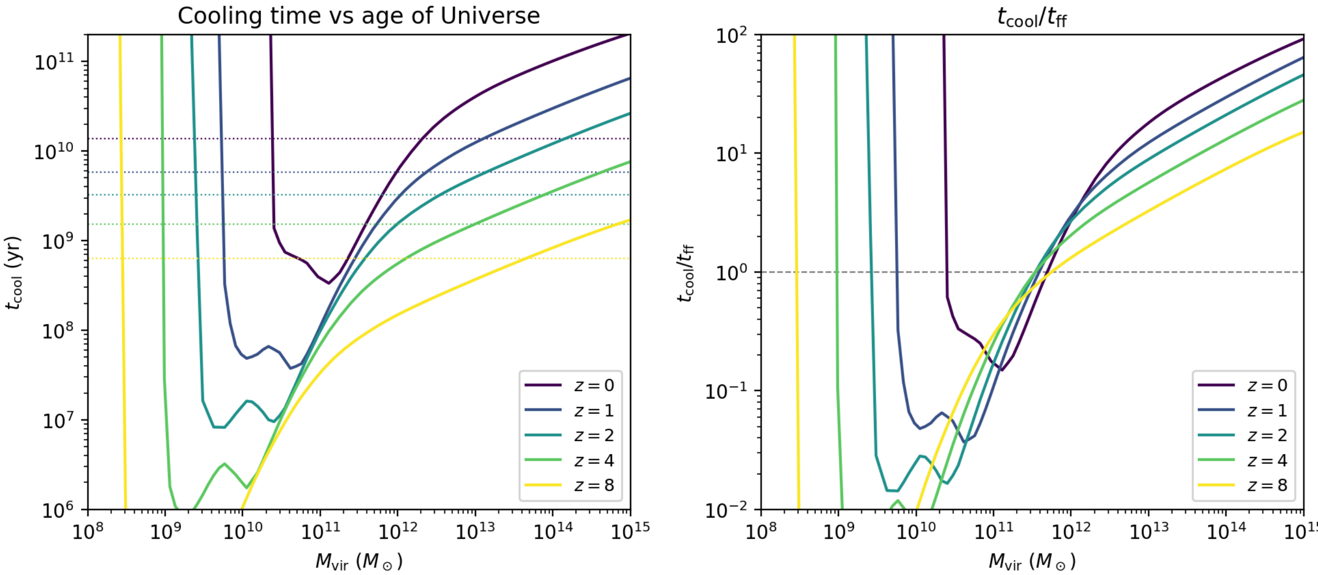
where Λ is the volumetric cooling rate (erg s^{-1} cm^{-3}).Since $\Lambda \propto n^2$, we have $t_{\rm cool} \propto 1/n$ — denser gas cools faster.The quantity $n \cdot t_{\rm cool}$ removes the density dependence and shows the intrinsic cooling efficiency at each temperature.



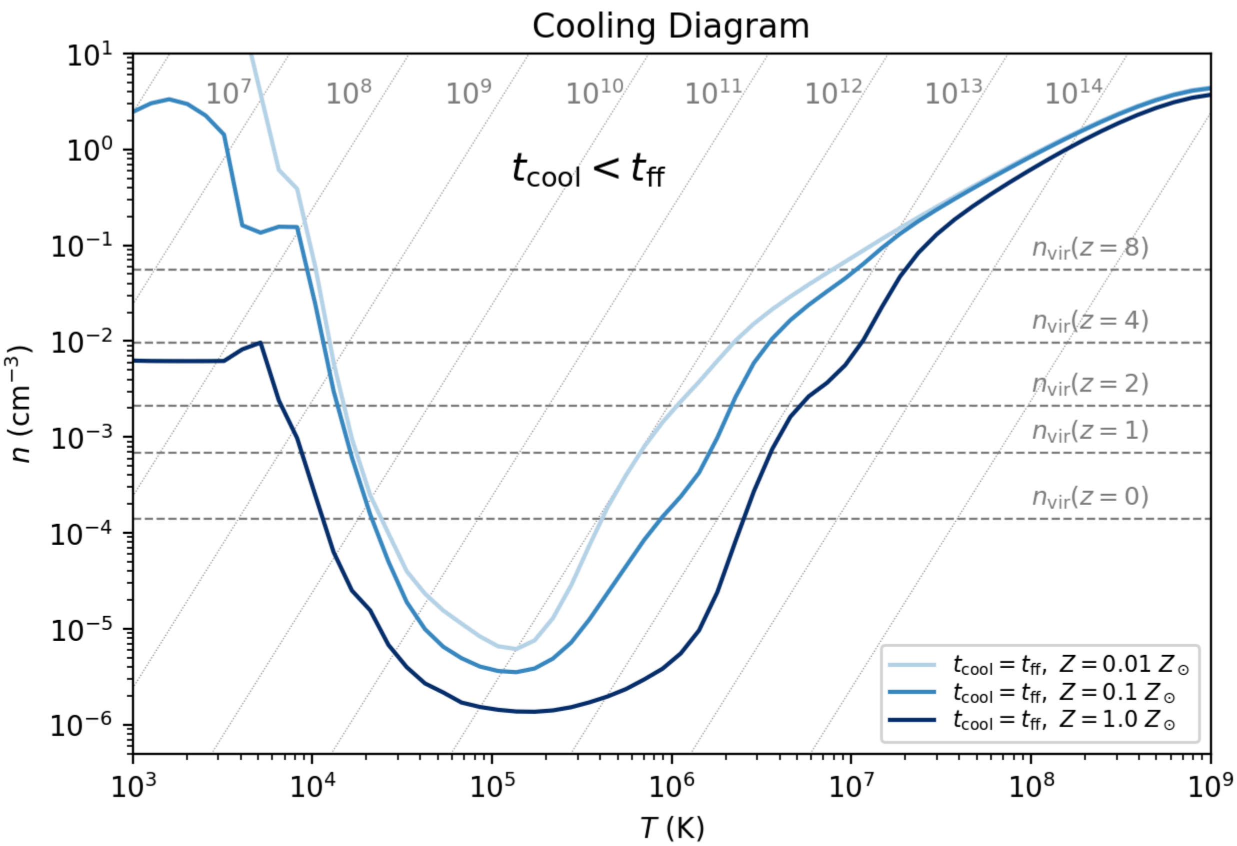
Exercise 5: Cooling Time vs. Free-Fall TimeThe free-fall time is the timescale for gravitational collapse:

$$t_{\rm ff} = \sqrt{\frac{3\pi}{32G\rho}}$$

The ratio $t_{\rm cool}/t_{\rm ff}$ determines the cooling regime:- $t_{\rm cool} < t_{\rm ff}$: rapid cooling — gas cools faster than it can fall, leading to fragmentation- $t_{\rm cool} > t_{\rm ff}$: slow cooling — gas settles into quasi-hydrostatic equilibrium (a hot gaseous halo)Compute $t_{\rm cool}$ and $t_{\rm ff}$ at the virial density for halos of mass 10^8–$10^{15}\ M_\odot$ at several redshifts. Compare to the age of the Universe.



The Cooling Diagram in the n-T PlaneWe can visualize the boundary between rapid and slow cooling in the density-temperature plane. The curve $t_{\rm cool} = t_{\rm ff}$ separates the two regimes:- Above the curve (high density): $t_{\rm cool} < t_{\rm ff}$ — cooling is efficient- Below the curve (low density): $t_{\rm cool} > t_{\rm ff}$ — cooling is slowWe also plot lines of constant halo mass. A halo at a given redshift sits at a specific point in this plane (determined by its virial temperature and virial density). As halos grow, they move along these mass tracks.



Reionization and the Baryon FractionAfter reionization ($z \sim 6$–10), the UV background heats the intergalactic medium to $\sim 10^4$ K. This prevents gas from accreting onto small halos whose virial temperature is below this threshold.Gnedin (2000) proposed a fitting function for the suppressed baryon fraction:

$$\frac{f_b}{\bar{f}_b} = \left[1 + (2^{\alpha/3} - 1)\left(\frac{M}{M_C}\right)^{-\alpha}\right]^{-3/\alpha}$$

where $\alpha = 2$ and $M_C(z)$ is a critical mass from Okamoto et al. (2008):

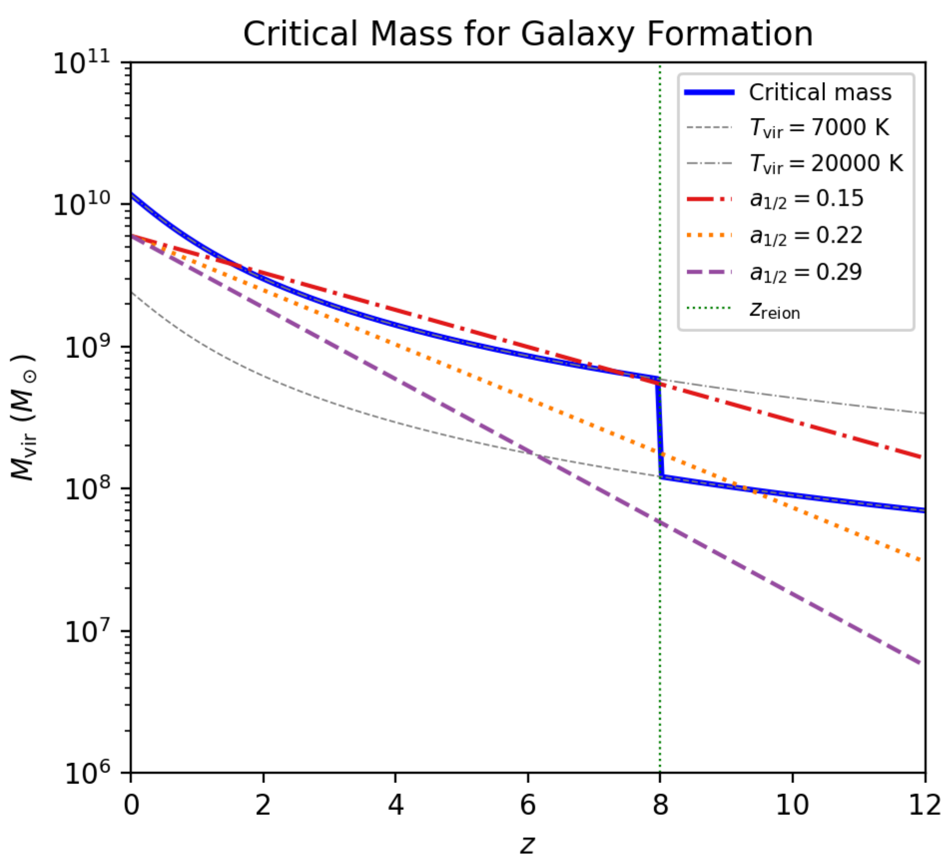
$$M_C(z) = \frac{6 \times 10^9}{h}\, e^{-0.63z}$$

Below M_C, the baryon fraction is strongly suppressed — these halos cannot accrete gas efficiently.

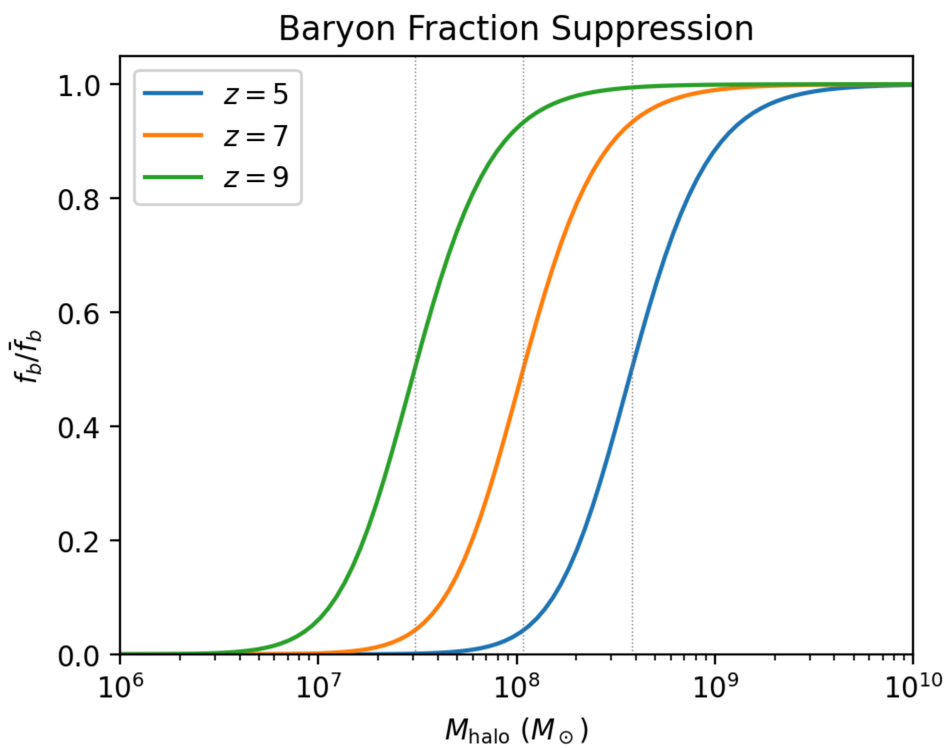
Exercise 6: Critical Mass for Galaxy FormationCombine two ideas to determine which halos can form stars:1. Before reionization ($z > z_{\rm reion}$): the critical mass is set by the atomic hydrogen cooling limit, $T_{\rm vir} \sim 7000$ K.2. After reionization ($z < z_{\rm reion}$): the UV background raises the threshold to $T_{\rm vir} \sim 20{,}000$ K.Overplot three idealized halo mass accretion histories (Wechsler+2002):

$$M(z) = M_0\, e^{-2a_{1/2}\, z}$$

with $M_0 = 6 \times 10^9\, M_\odot$ and different formation parameters $a_{1/2}$. Determine which halos can form stars.



Baryon Fraction SuppressionThe plot below shows the suppressed baryon fraction as a function of halo mass at different redshifts. Small halos ($M \ll M_C$) lose most of their baryons after reionization.



Demonstration: Combined Critical Mass Diagram

The figure below combines the two key constraints on galaxy formation into a single diagram:

1. M_{c1} **(Reionization)**: Below this mass, halos cannot accrete baryons due to the UV background. Before reionization, the limit is set by atomic cooling ($T_{\rm vir} \sim 7000$ K); after reionization, it jumps to $T_{\rm vir} \sim 20{,}000$ K (Okamoto et al. 2008).

2. M_{c2} **(Cooling)**: Above this mass, the cooling time exceeds the age of the Universe — gas cannot cool and collapse. This sets an upper mass limit for efficient galaxy formation.

The gray shaded region shows the range of halo masses that actually exist at each redshift, defined by the peak height range $0.5 < \nu < 5$. Galaxies can only form in the window between M_{c1} and M_{c2} that overlaps with this region (adapted from CFN Chapter 8).

Summary

Part 1 -- The Galaxy-Halo Connection:

1. Abundance matching connects galaxies to halos by matching cumulative number densities. The most massive galaxy lives in the most massive halo.
2. The stellar-to-halo mass relation (SHMR) shows that star formation efficiency peaks at $M_{\rm halo} \sim 10^{12}\,M_\odot$ (Milky Way mass), where $\sim 20\%$ of available baryons become stars. Supernova feedback suppresses efficiency in low-mass halos; AGN feedback suppresses it in high-mass halos.
3. Galaxies are biased tracers of the dark matter distribution: $P_{gg}(k) = b^2 P_{mm}(k)$. Massive halos are more biased ($b > 1$). The effective bias of a galaxy sample depends on the halo mass threshold.

Part 2 -- Gas Accretion & Cooling: 4. Gas shock-heats to $T_{\rm vir} \propto M^{2/3}$ when accreting onto halos. MW-mass halos have $T_{\rm vir} \sim 4 \times 10^5$ K. 5. The net cooling rate depends on T, n, and Z. Metal-line cooling dominates at 10^4--10^7 K; bremsstrahlung dominates above 10^7 K. 6. When $t_{\rm cool} < t_{\rm ff}$, gas cools rapidly and fragments. When $t_{\rm cool} > t_{\rm ff}$, a hot gaseous halo forms. Massive clusters have $t_{\rm cool} > t_{\rm age}$, so their gas never fully cools. 7. Reionization suppresses gas accretion onto halos with $T_{\rm vir} < 2 \times 10^4$ K ($M \lesssim 10^{9.5}\,M_\odot$), setting a minimum halo mass for galaxy formation.